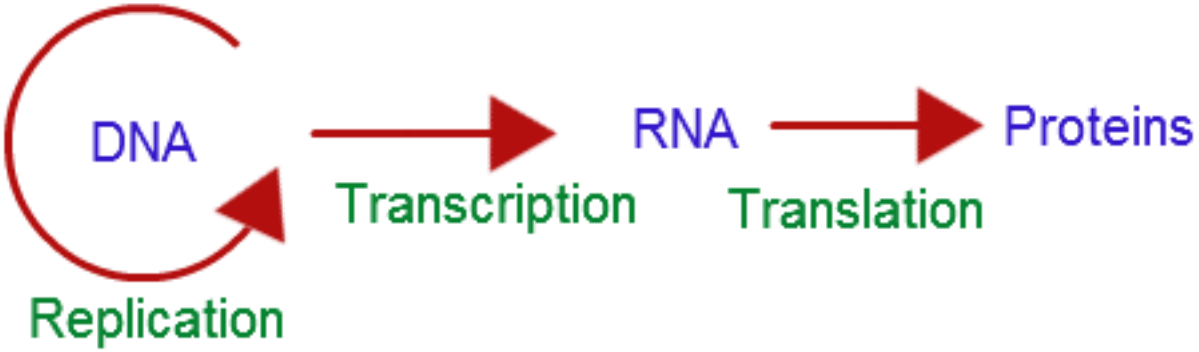
Gene Expression

Step 1: Transcription

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Central Dogma: The Flow of Genetic Information



Transcription refers to RNA synthesis using DNA as a template

Translation is the synthesis of protein using the information in the RNA

Why is there such a thing as RNA?

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Transcription and Translation Involve Many of the Same Components in Bacteria and Eukaryotes

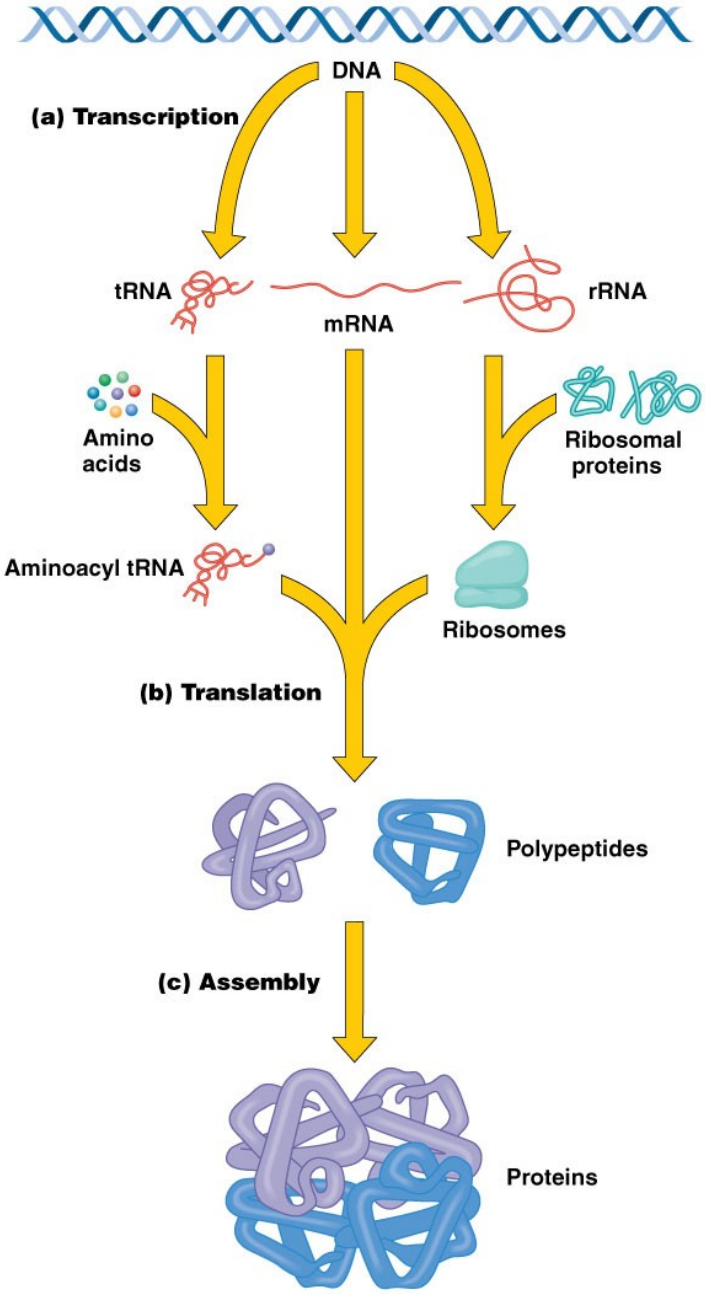
Messenger RNA, mRNA, is RNA that is translated into protein

Ribosomal RNA, rRNA, is an integral component of the ribosome

Transfer RNA, tRNA, molecules serve as intermediaries, bringing amino acids to the ribosome

The latter two function in translation

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All three major classes of RNA—tRNA, mRNA, and rRNA—are

(a) synthesized by transcription of the appropriate DNA sequences (genes) and

(b) involved in the subsequent process of translation (polypeptide synthesis). The appropriate amino acids are brought to the mRNA and ribosome by tRNA. A tRNA molecule carrying an amino acid is called an aminoacyl tRNA.

(c) Polypeptides then fold and assemble into functional proteins.

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Genetic Code

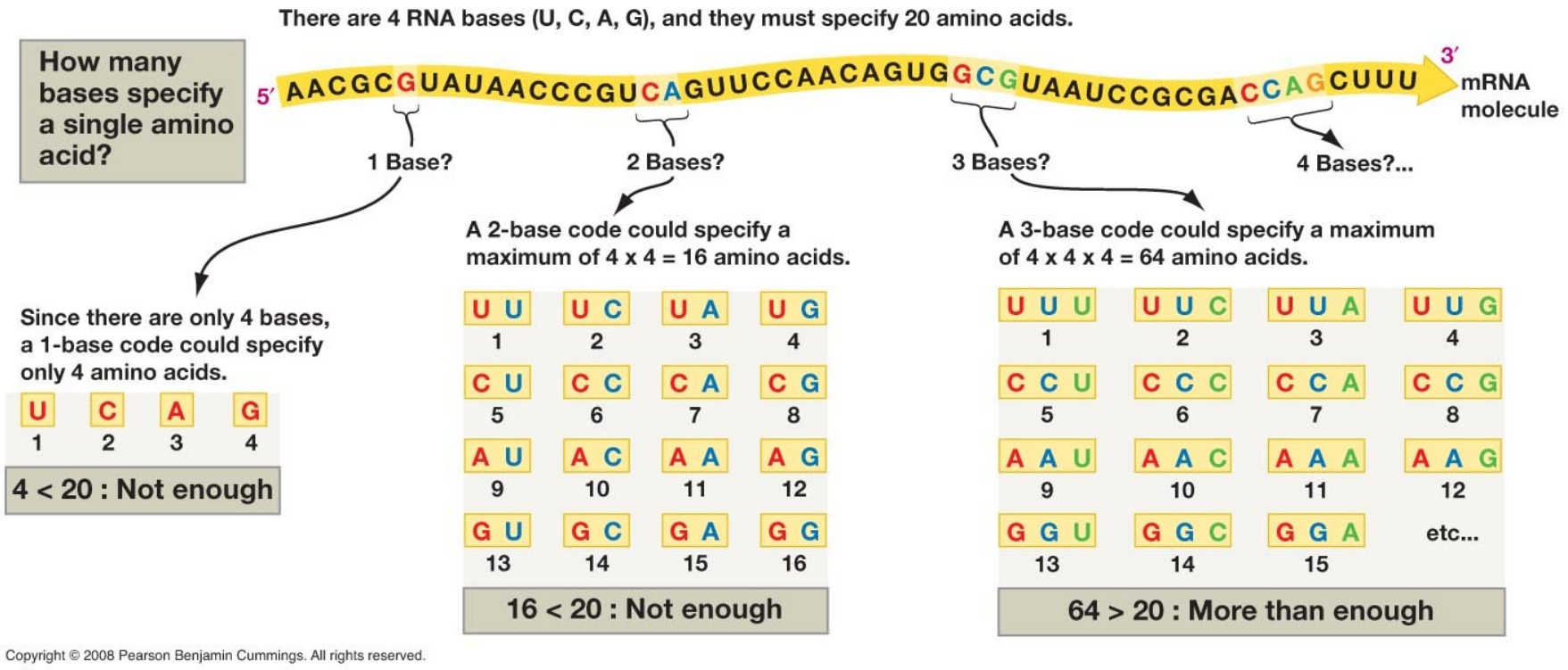
- How does RNA determine the order of amino acids in a protein?
- The relationship between the DNA base sequence and the linear order of amino acids in the protein products is based on a set of rules known as the **genetic code.**

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The Genetic Code

- **Genes** are defined as functional units of DNA that encode one or more polypeptides or functional RNA molecule. (Genes don't always have to code for proteins!)
- How many nucleotides in DNA are needed to specify each amino acid in a protein? → Triplet code
- A code in which three base pairs in double-stranded DNA are required to specify each amino acid in a polypeptide.

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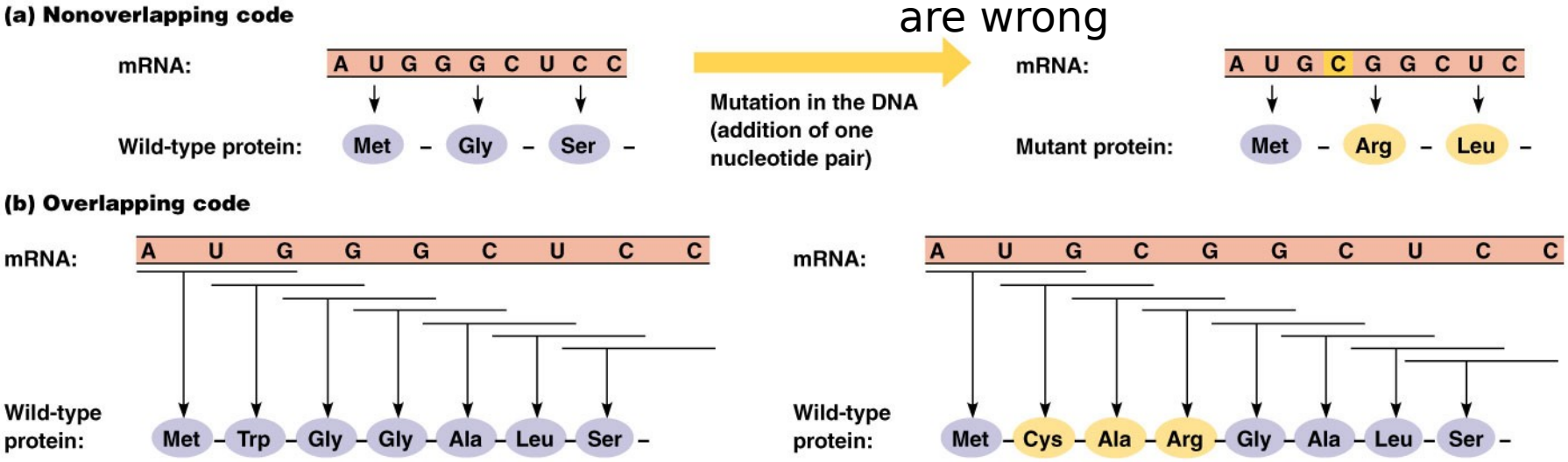
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The Genetic Code Is Degenerate and Nonoverlapping

- There are 64 (4x4x4 possibilities coming from 4 nts) combinations of nucleotide triplets and only 20 amino acids
- This means the genetic code is a **degenerate code,** meaning that a particular amino acid can be specified by more than one nucleotide triplet
- It is also *nonoverlapping*; the reading frame advances three nucleotides at a time

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insertion alters the reading frame beyond that point, so the remainder of the mRNA is read incorrectly and all amino acids are wrong

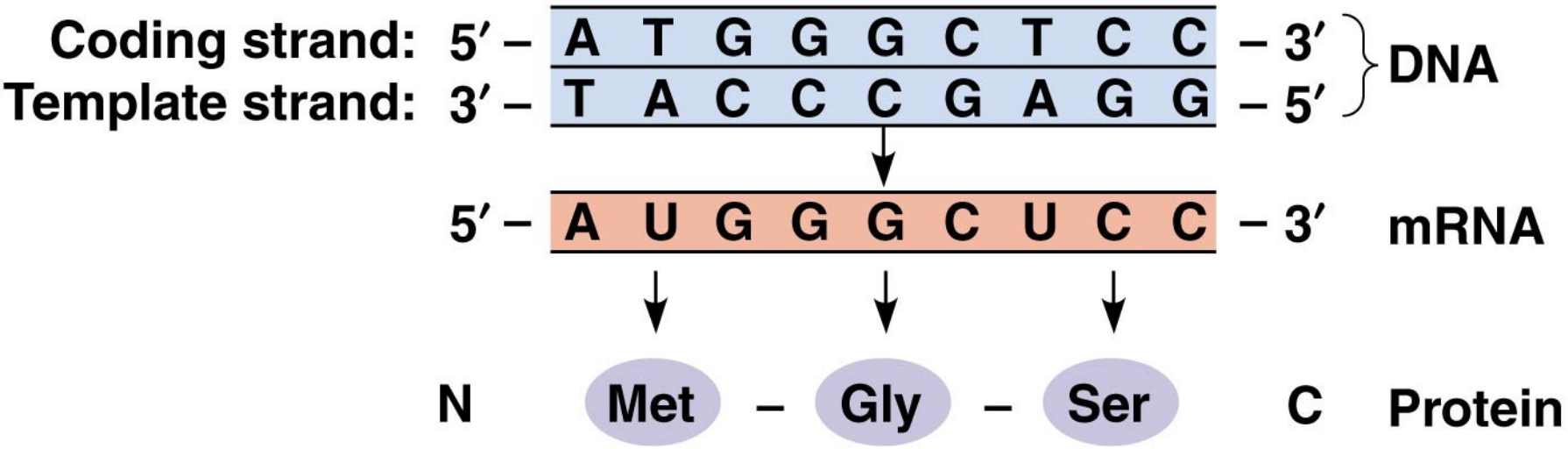


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Messenger RNA Guides the Synthesis of Polypeptide Chains

- **The genetic code** refers to the order of nucleotides in the mRNA molecules that direct protein synthesis
- In mRNA synthesis, only one DNA strand is copied, called the **template strand;** the other strand is called the **coding strand** because it is similar to the mRNA sequence
- In mRNA synthesis, an uracil base (U) is used instead of thymine (T).

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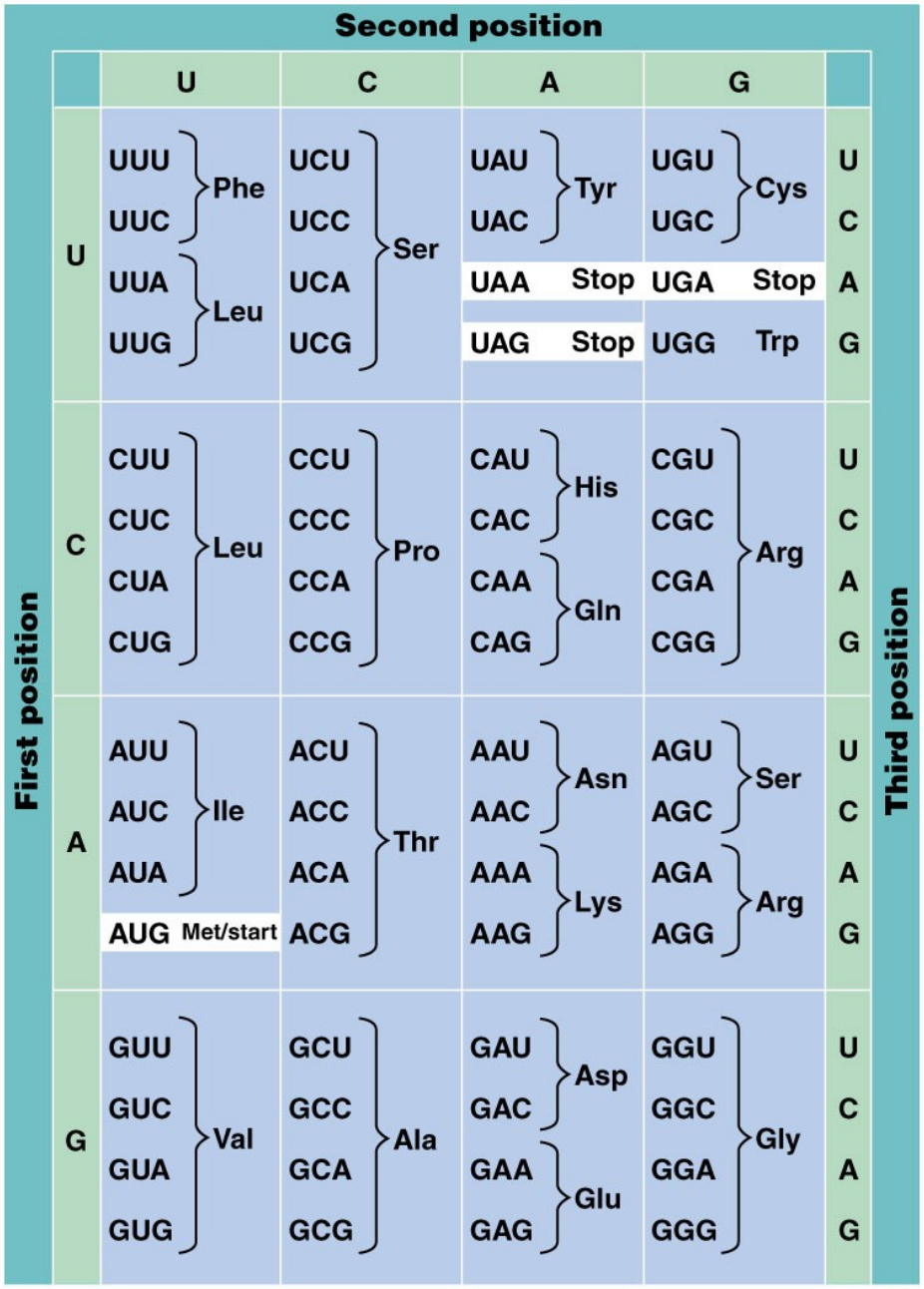
Directionality!

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Of the 64 Possible Codons in Messenger RNA, 61 Code for Amino Acids

- All 64 codons are used in the translation of mRNA
- 61 of them specify the addition of specific amino acids to a growing polypeptide chain
- One of them, AUG, plays a role as a **start codon**
- The remaining three (UAA, UAG, UGA) are **stop codons,** which terminate polypeptide synthesis

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The Genetic Code Is Unambiguous and Degenerate

- Every codon has one meaning only—the genetic code is *unambiguous*

- It is also *degenerate*—many of the amino acids are specified by more than one codon

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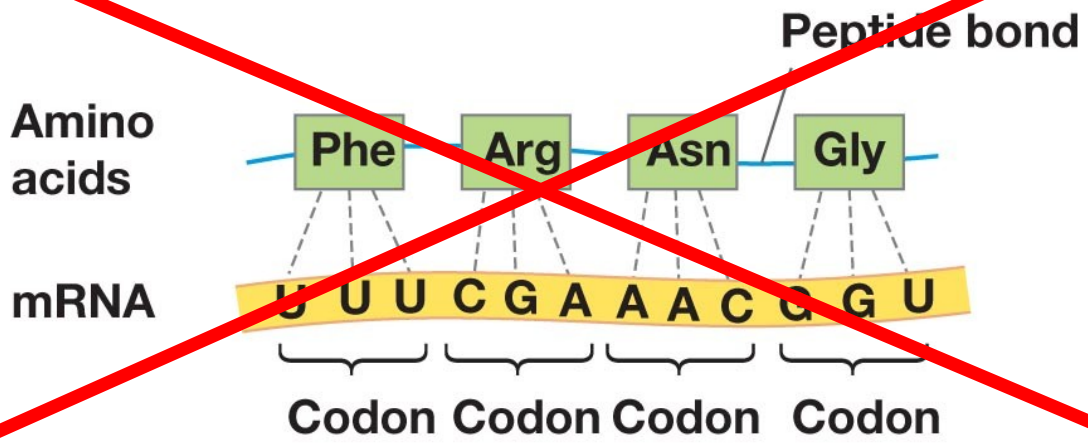
The Genetic Code Is (Nearly) Universal

- Except for a few cases, all organisms use the same basic genetic code

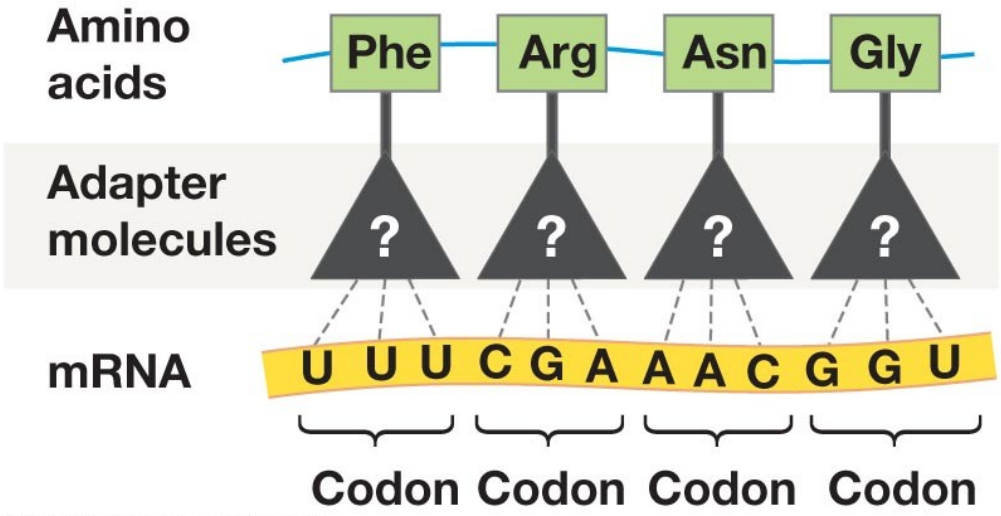
- In the case of mitochondria and a few bacteria, the genetic code differs in several ways
- For example, AGA is a stop codon in mammalian mitochondria, and in some organisms codons specify nonstandard (modified) amino acids which are not part of mammalian polypeptide chains.

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(a) Hypothesis 1: Amino acids interact directly with mRNA codons.



(b) Hypothesis 2: Adapter molecules hold amino acids and interact with mRNA codons.



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Mutations alter the genetic code

- One feature of genetic code is that it allows genetic variation in population so that it allows speciation and evolutionary process (Dynamic)

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Mutations and Translation

- mRNAs may contain mutant codons that cause errors in the polypeptide chain synthesized
- There are several types of mutations in which the DNA change involves only one or a few base pairs.
- A single base change can create a devastating genetic disorder or a beneficial adaptation, or it might have no effect.

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Missense Mutations

- In sickle-cell anemia, an A-T base pair was substituted for a T-A base pair in DNA
- As a result, a GUA codon replaces a GAA codon in mRNA transcribed from the mutant allele
- In polypeptide, this leads to a valine (neutral) replacing the normal glutamic acid (negatively charged) in the protein
- **This is called a missense mutation (wrong aminoacid)**

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Nonstop and Nonsense Mutations

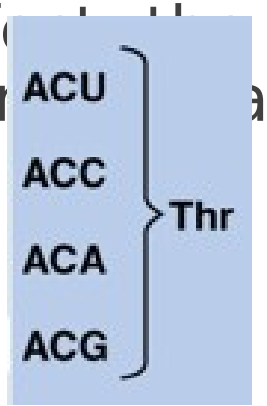
- A base-pair substitution can turn a stop codon to an amino acid codon → This is called a **nonstop mutation**

- Conversely, the mutation can change an amino acid codon to a stop codon → This is called a **nonsense mutation** and can lead to premature termination of translation and a shorter polypeptide than normal

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Other Types of Mutations

- Sometimes a single amino acid change may not affect a protein's function in a major way
- Substituting one amino acid for another may not affect the protein function if both amino acids are similar
- Some mutations may be ***silent mutations*** that affect the third base of the codon and results in no change of amino acid



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Indels

- Nonsense, nonstop, and missense codons can also arise from *base-pair insertions, deletions,* or a combination of these

- These are called *indels*, and they cause *frameshift mutations*

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Frameshift Mutations

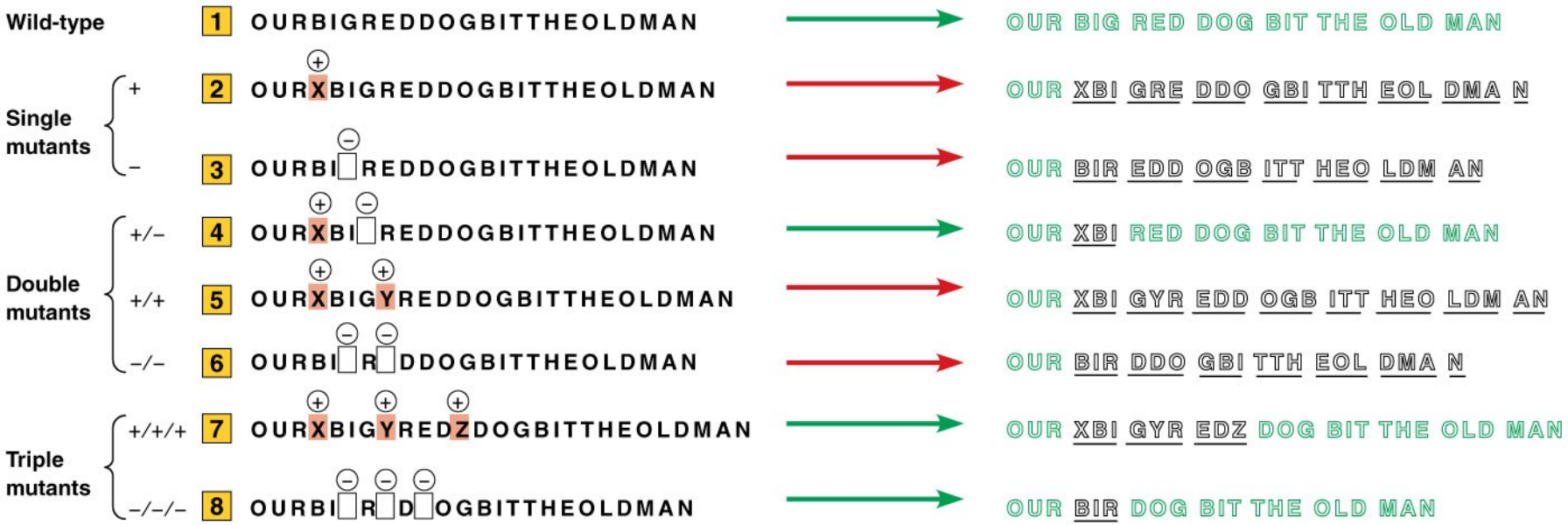
- The gene is written in a language of three-letter words (triplet code)
- Inserting or deleting a nucleotide (**indel** mutations) causes the rest of the sequence to be read out of phase—this is a shift in the *reading frame*
- Mutations that cause insertion or deletion of a nucleotide are thus called **frameshift mutations**

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Larger-Scale Mutations

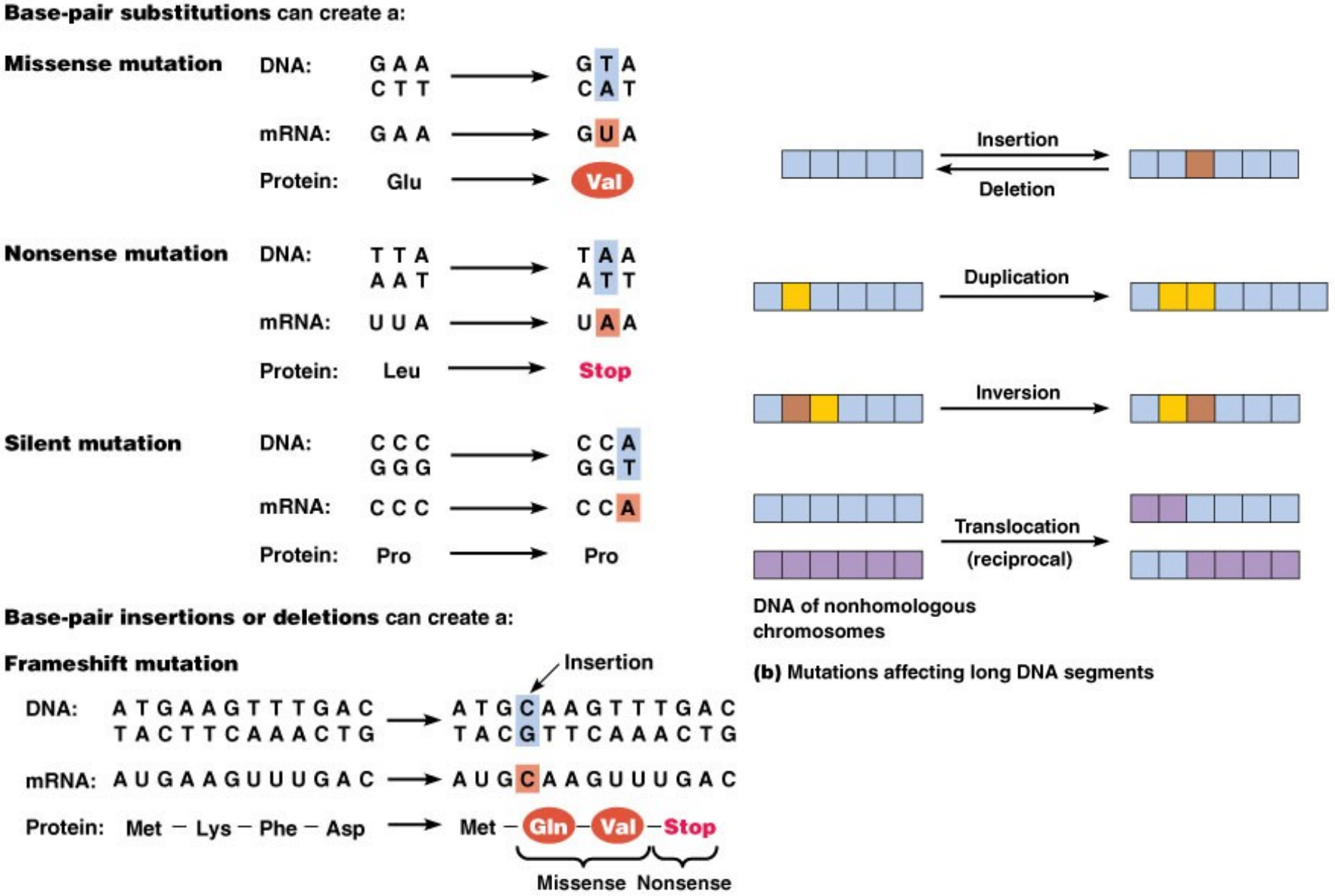
- Some larger-scale mutations are created by *indels* involving longer segments of DNA
- In *duplications,* a stretch of DNA is repeated
- In an *inversion,* a chromosome section is cut out and reinserted in its original position but in reverse orientation
- A *translocation* involves movement of a segment from one position in the genome to another location

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The frameshift mutations would not have been observed if the genetic code were overlapping. Accordingly, their data indicated the code to be nonoverlapping.

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Sickle Cell Anemia

Sickle cell disease is caused by a single mutation in hemoglobin DNA sequence that results in a change in hemoglobin aminoacid sequence.

https://www.dnalc.org/resources/3d/17-sickle-cell.html

The normal hemoglobin mRNA sequence is below:

5’GUGCACCUGACUCCUGAGGAG3’

What is the coding sequence of DNA?

What is the template strand sequence of DNA?

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Sickle Cell Anemia

The sickle cell hemoglobin DNA sequence is below.

The bottom strand is the template strand and the top strand is the coding strand.

5’GTGCACCTGACTCCTGAGGTG3’

3’CACGTGGACTGAGGACTCCAC5’

What is the sequence of mRNA 5’□3’?

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Sickle Cell Anemia

Let’s compare the sequences of mRNA:

HbA (normal) 5’GUGCACCUGACUCCUGAGGAG3’

HbS (sickle) 5’GUGCACCUGACUCCUGUGGAG3’

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Sickle Cell Anemia

Now let’s convert normal and sickle cell hemoglobin mRNA into peptide:

HbA (Normal):

5’GUG CAC CUG ACU CCU GAG GAG3’

HbS (sickle):

5’GUG CAC CUG ACU CCU GUG GAG3’

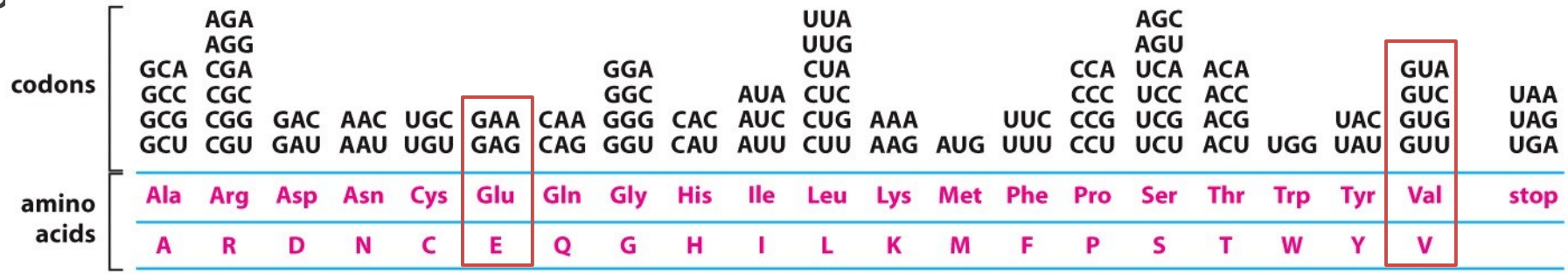


Figure 7-25 Essential Cell Biology, 4th ed. (© Garland Science 2014)

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Sickle Cell Anemia

HbA (Normal):

N-Val-His-Leu-Thr-Pro-Glu-Glu-C

HbS (sickle):

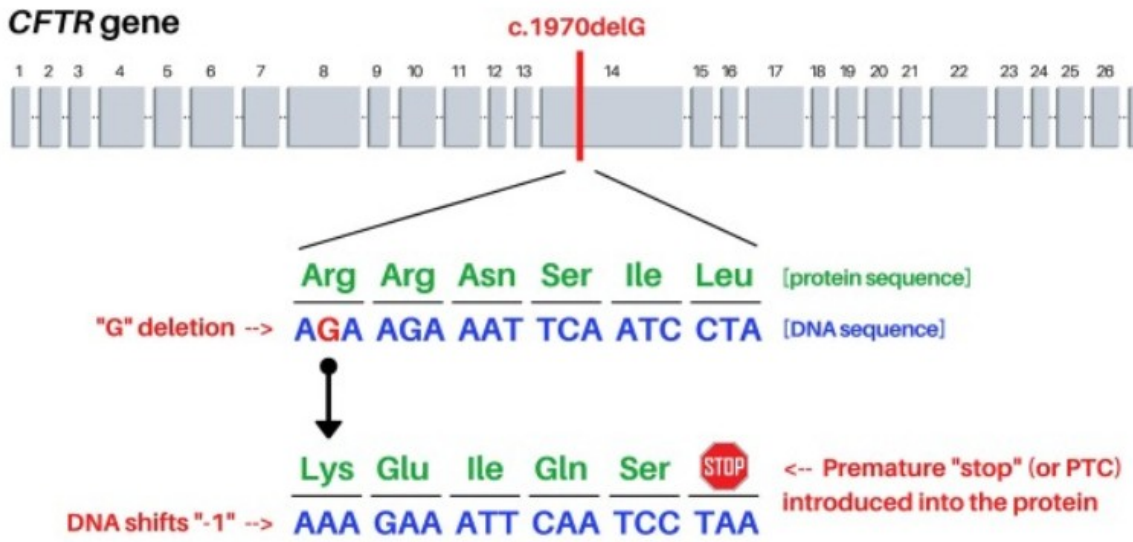
N-Val-His-Leu-Thr-Pro-Val-Glu-C



Sickle cell disease is caused by a single mutation in hemoglobin DNA sequence that results in a change in hemoglobin aminoacid sequence.

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Cystic Fibrosis

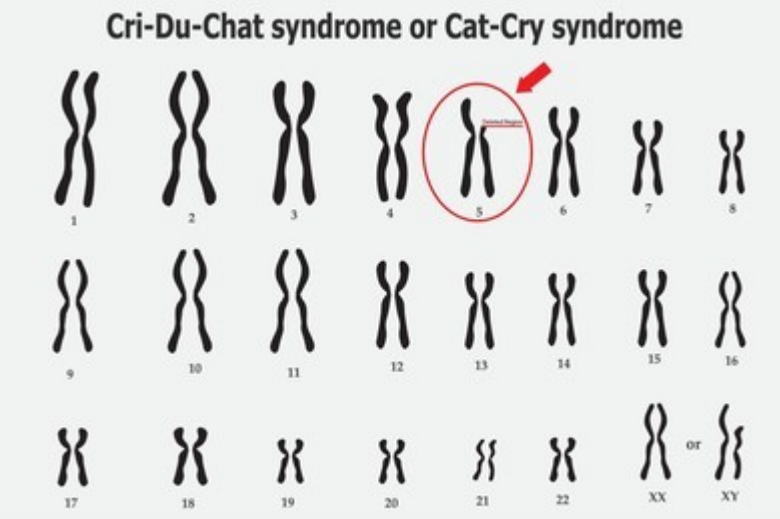


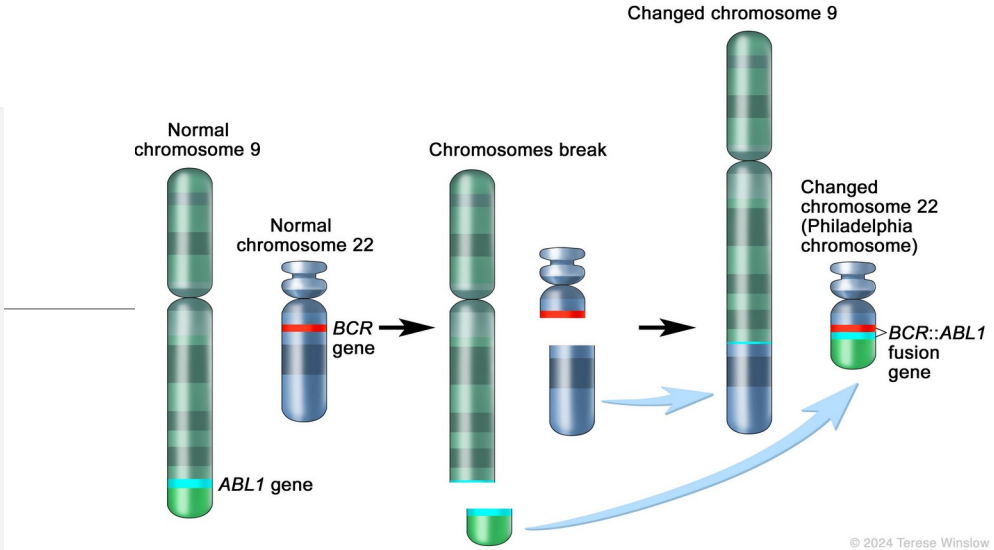
The mRNA is recognized as aberrant and degraded

The protein is truncated in half, degraded by the proteasome.

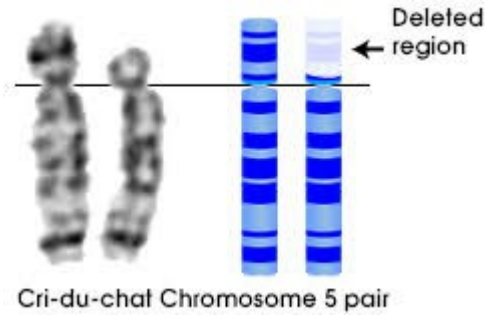
Overall: NO CTFR functional expression.

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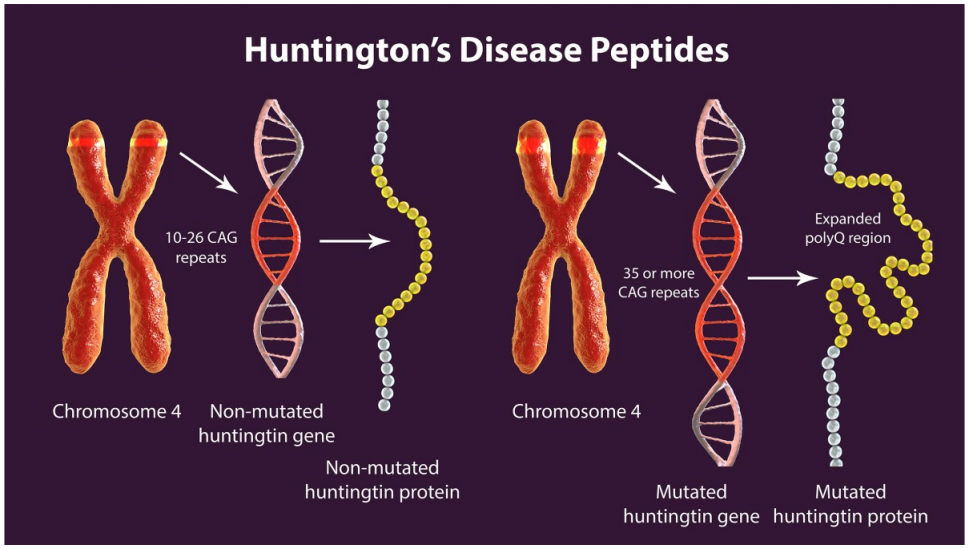




translocation



11.2 megabases (Mb) to **28.6 Mb**



trinucleotide repeat expansion

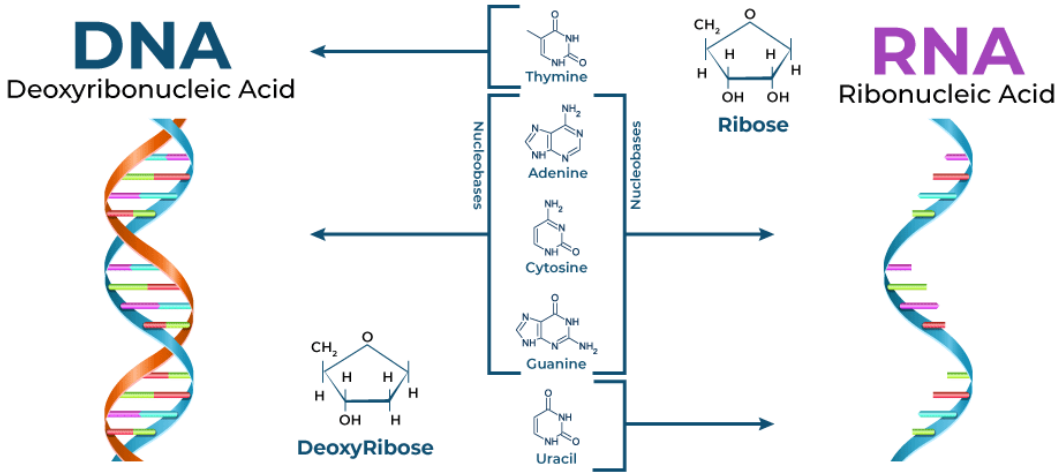
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Transcription

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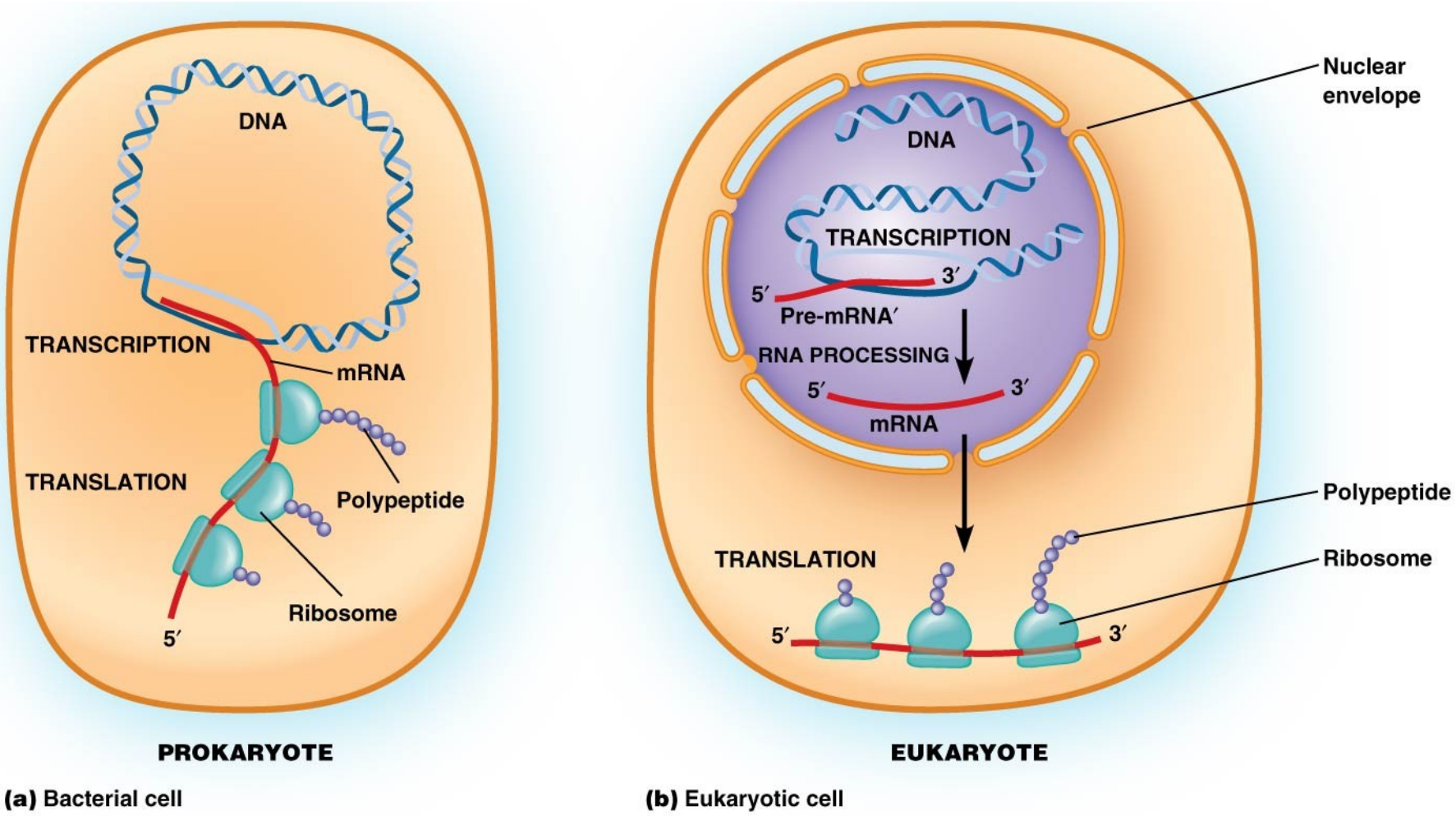
18.2 Mechanisms of Transcription

- RNA is chemically similar to DNA but contains ribose instead of deoxyribose
- It also has the base uracil instead of thymine and is usually single stranded



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- The fundamental principles of transcription were first elucidated in bacteria



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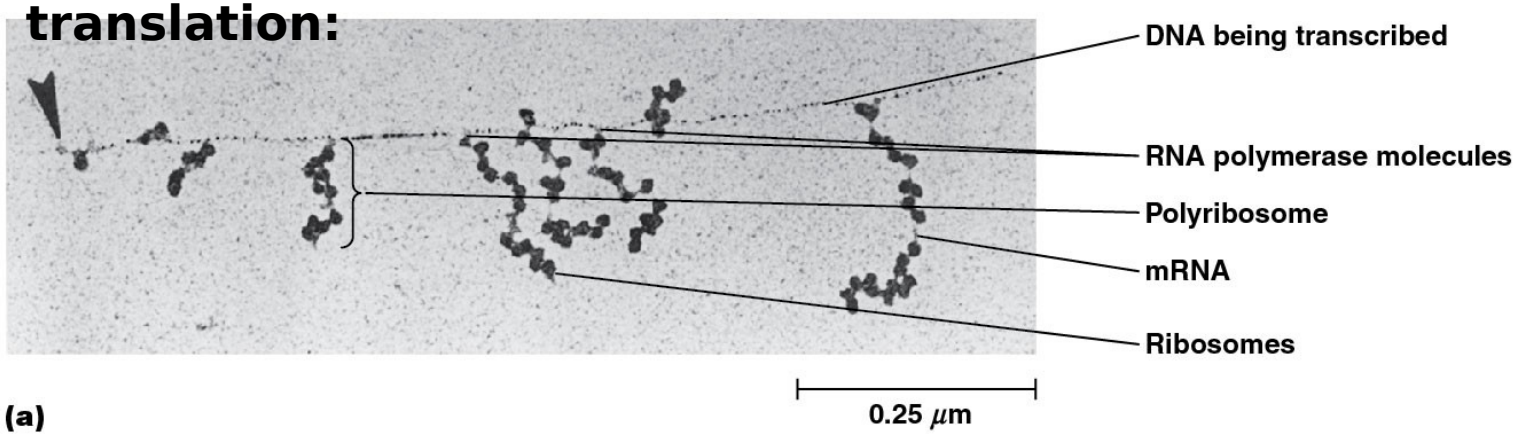
Where Transcription and Translation Occur Differs in Bacteria and Eukaryotes

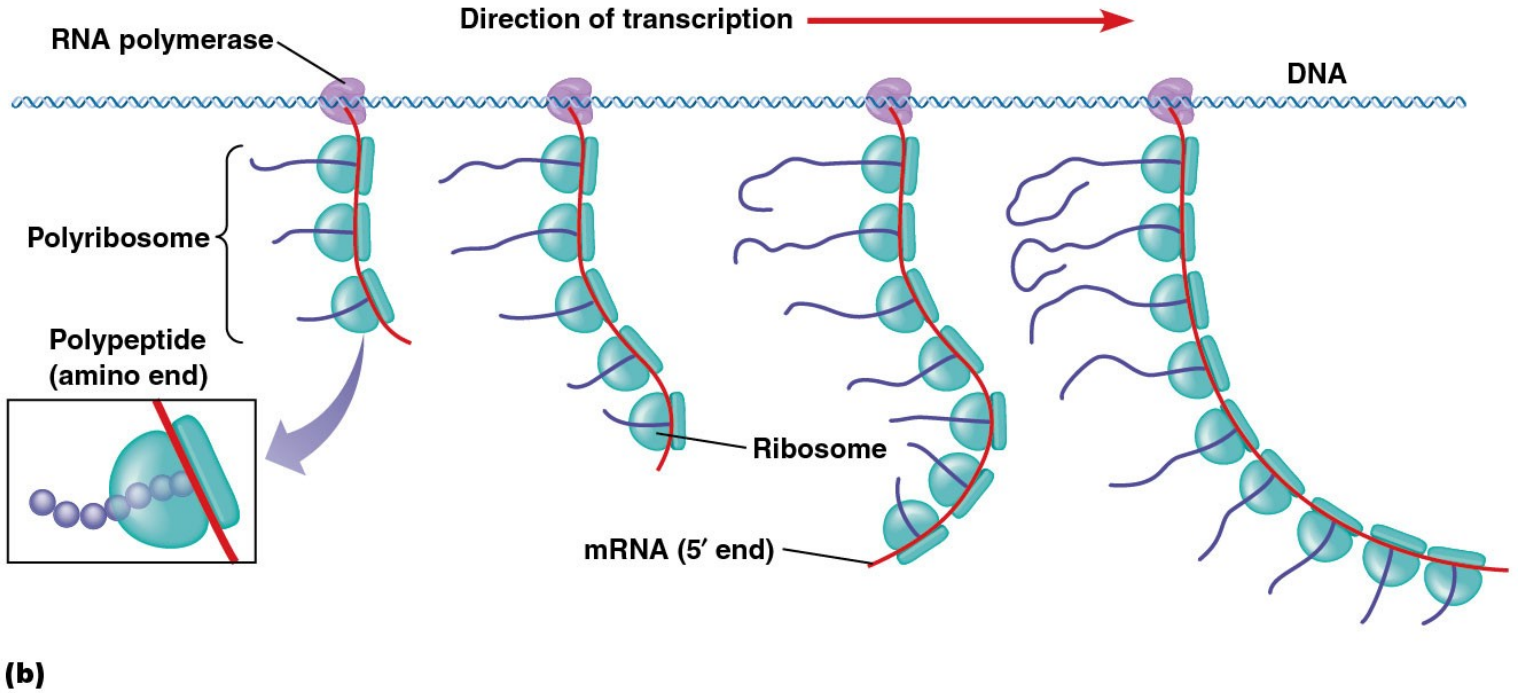
- Because bacteria do not have a nuclear envelope, **translation of mRNA can begin before its transcription is completed** → quick and higher yield
- The compartmentalization of eukaryotic cells leads to **spatial separation of transcription and translation**

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The coupling of transcription with translation:

Arrow: Txn start site





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Transcription Involves Four Stages: Binding, Initiation, Elongation, and Termination

- The DNA that gives rise to one RNA molecule is called the **transcription unit=gene**

- Transcription begins when RNA polymerase binds to a promoter sequence (1), triggering local unwinding of the double helix
  - **No DNA helicase is involved in transcription**, only in DNA replication.
    Transcription and DNA replication are two fundamentally differ ent processes. DNA replication requires unwinding the whole g enome and synthesising it again! Transcription involves creatin g RNA from one strand of DNA in a single gene

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Steps of RNA Synthesis (continued)

- RNA polymerase then initiates synthesis of RNA using one DNA strand as a template (2)
- After initiation, the RNA polymerase moves along the DNA template, unwinding the helix and elongating the RNA (3)
- The unwound DNA is called a *transcription bubble*
- Eventually, RNA polymerase dissociates from the DNA template (4), leading to termination of synthesis and release of the RNA molecule

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Let’s review strand specificity and sequences of DNA and RNA

- Template strand of DNA is the strand RNA Pol II reads. Pol II moves in 3’→5’ direction on this strand.
- Coding strand of DNA is the strand that has the same sequence as the mRNA molecule (Uracil in RNA).
- RNA pol II synthesizes mRNAs in 5’→3’ direction

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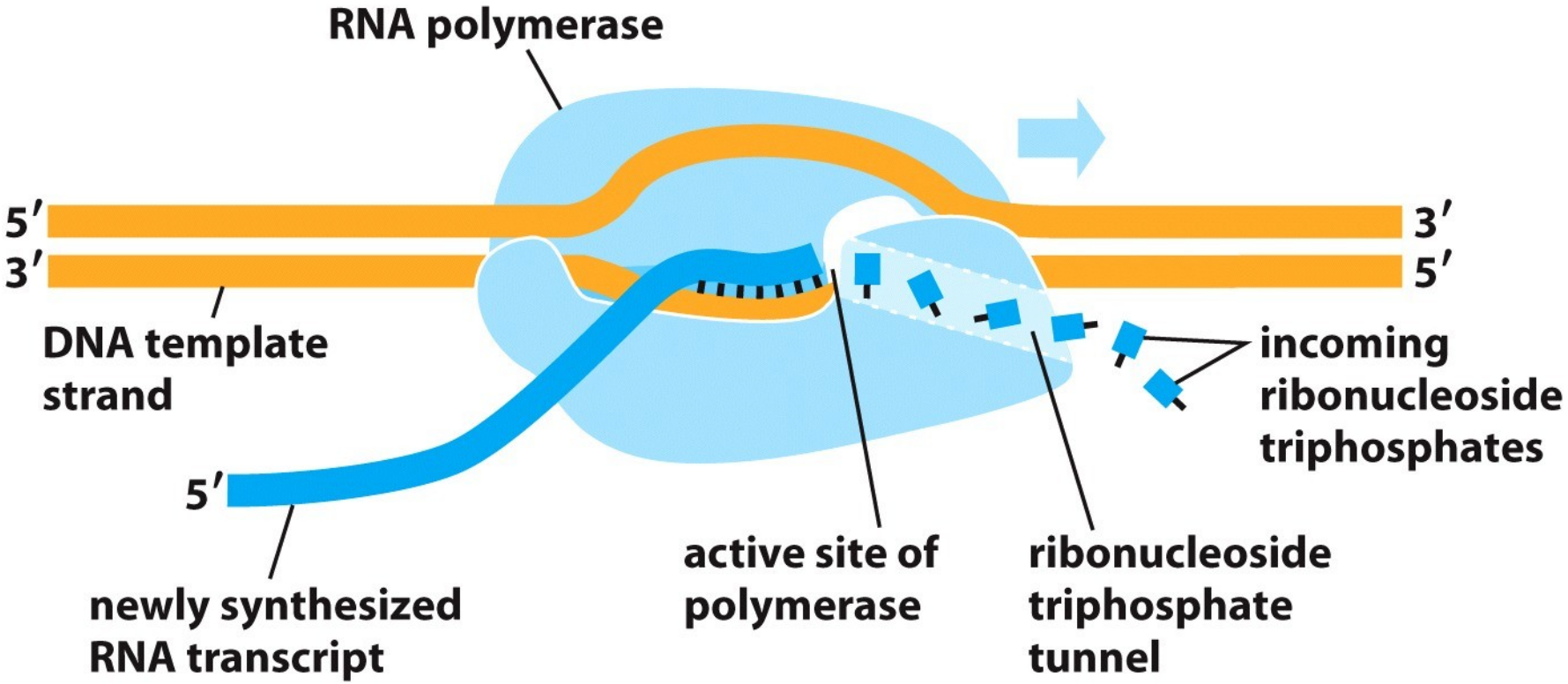
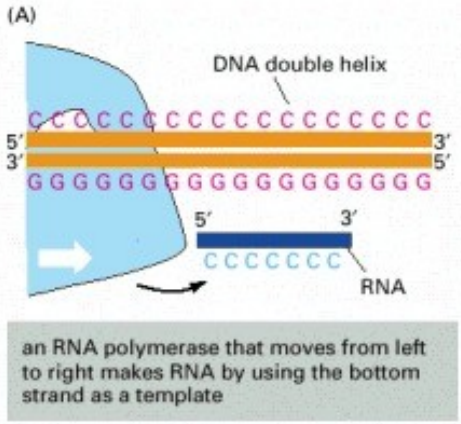


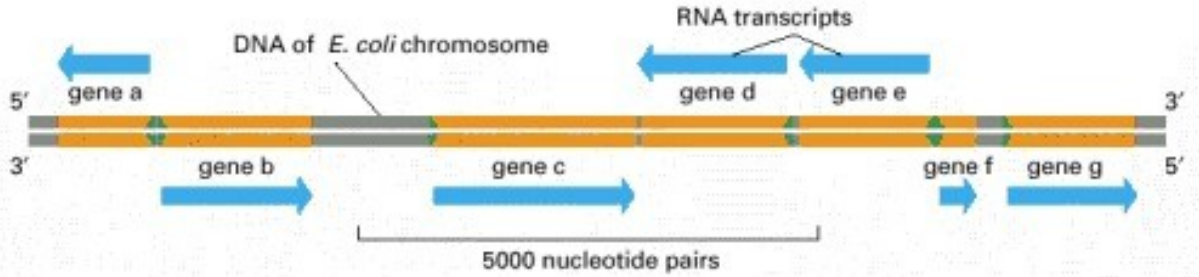
Figure 7-7 Essential Cell Biology, 4th ed. (© Garland Science 2014)

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- The choice of template strand for each gene is determined by the location and orientation of the promoter.

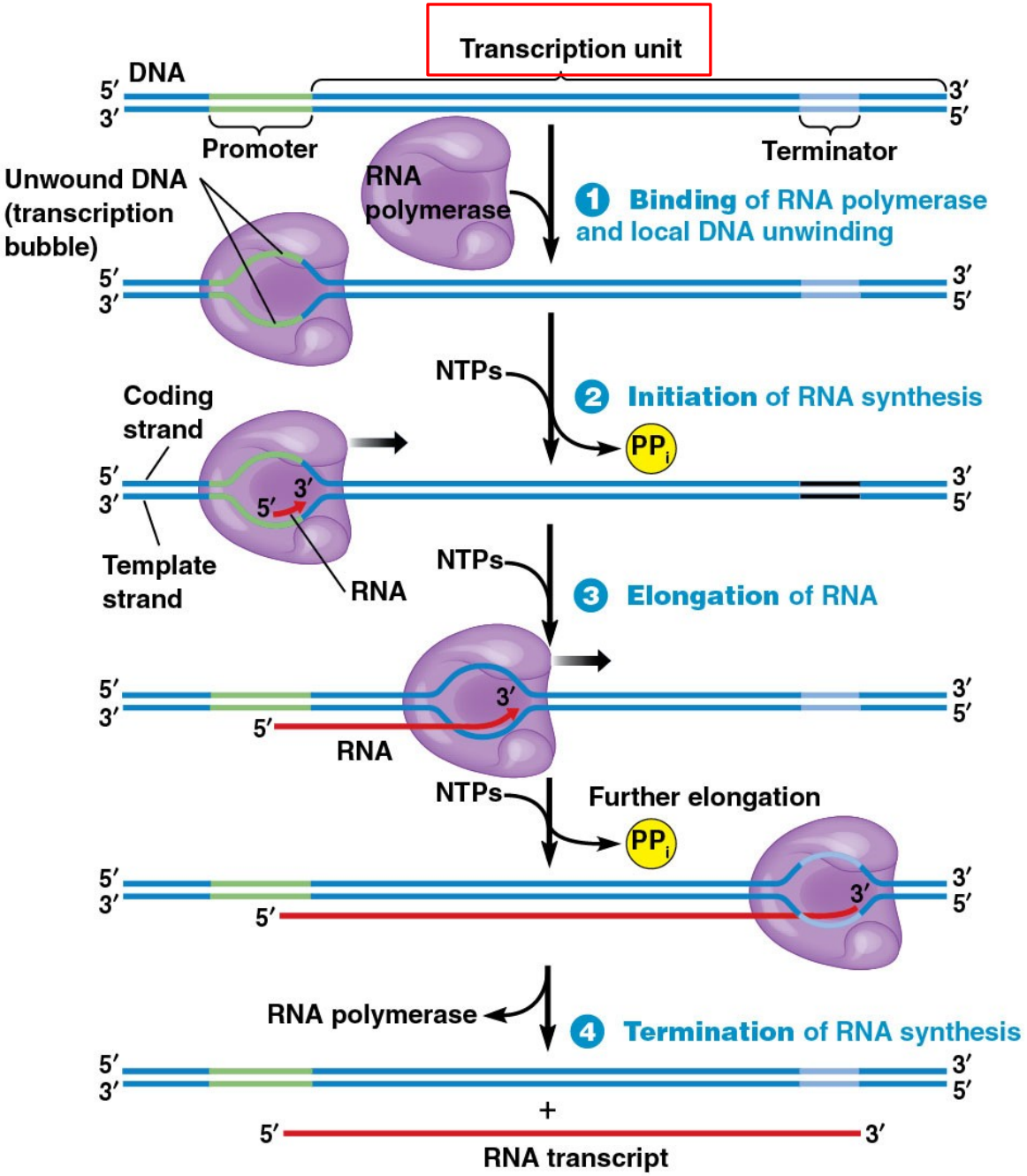


The DNA strand serving as template must be traversed in a 3′ to 5′ direction. Thus, the direction of RNA polymerase movement determines which of the two DNA strands is to serve as a template for the synthesis of RNA, as shown in (A) and (B).



DNA strand used as the template for RNA synthesis varies from gene to gene!

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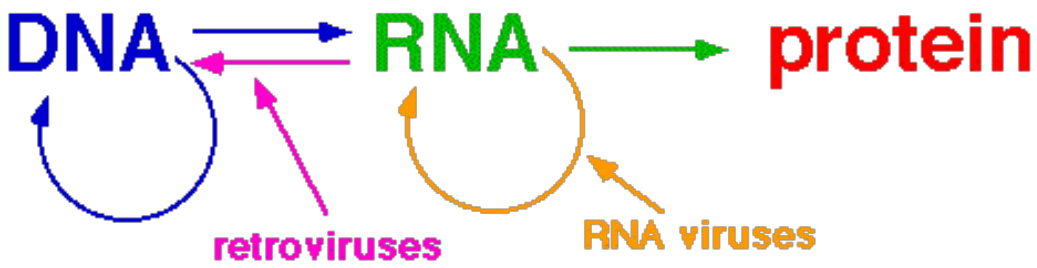


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Exceptions of the central dogma: Retroviruses

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Refinements of the Central Dogma



- There are exceptions to the central dogma
- For example, there are RNA viruses that carry out *reverse transcription,* using RNA as a template for DNA synthesis
- Other viruses produce RNAs from an RNA template

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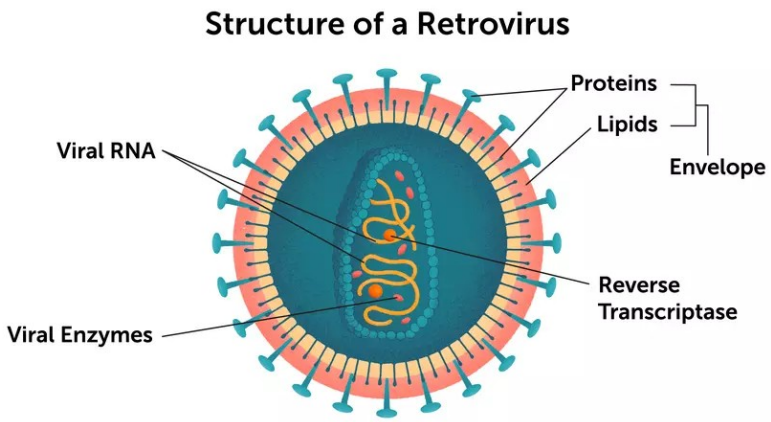
In Some Cases RNA Is Reverse Transcribed into DNA

- In certain cases, RNA can serve as a template for DNA synthesis
- This process of *reverse transcription* is catalyzed by the enzyme **reverse transcriptase**
- Viruses that carry out reverse transcription are called **retroviruses**

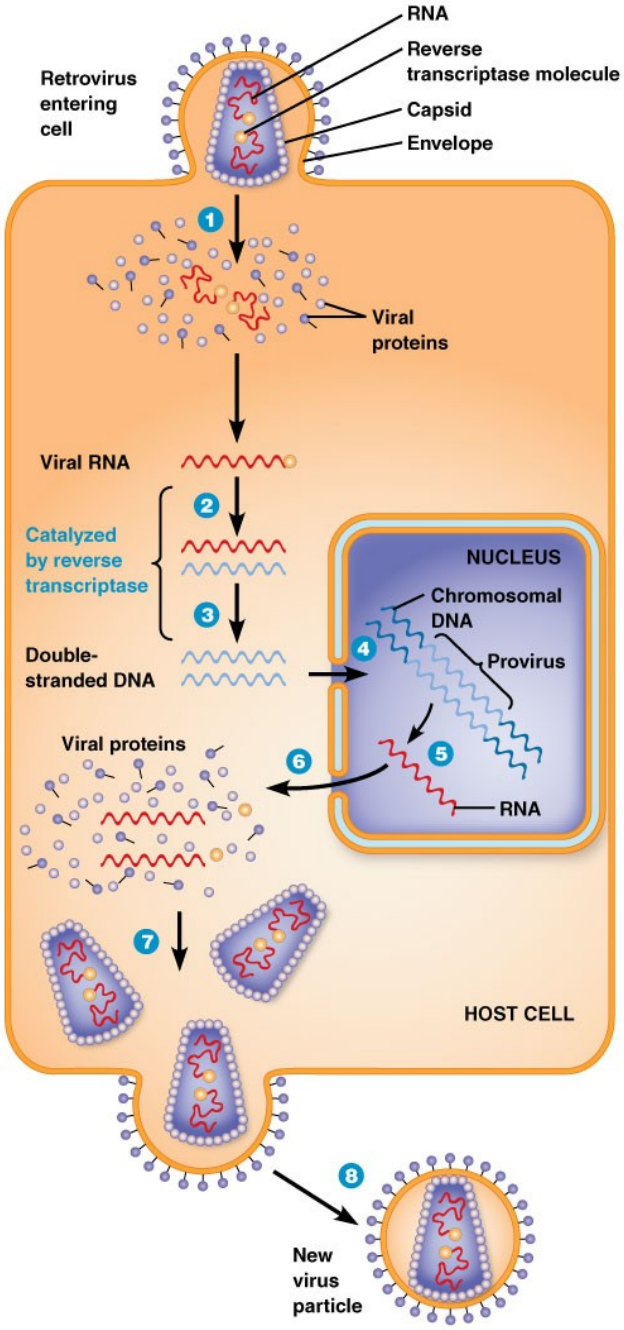
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Examples of Retroviruses

- *Human immunodeficiency virus* (*HIV*) causes *acquired immune deficiency syndrome* (*AIDS*)
- Coronavirus
- A numb se cancers in anima



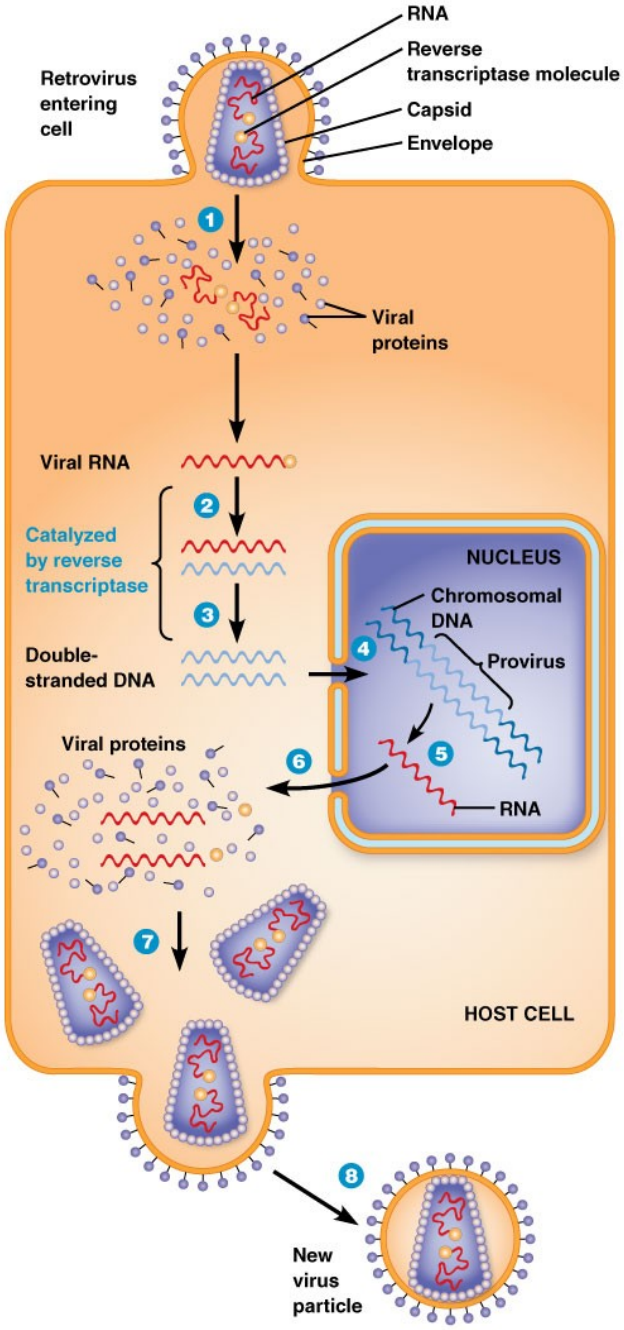
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Transcription of Retroviral DNA

- In the virus, two copies of the RNA genome are enclosed within a protein capsid
- Each RNA molecule has an attached copy of reverse transcriptase
- The virus first binds the surface of the host cell, and its envelope fuses with the plasma membrane
- Once inside the cell, the viral reverse transcriptase catalyzes synthesis of a DNA strand complementary to the viral RNA

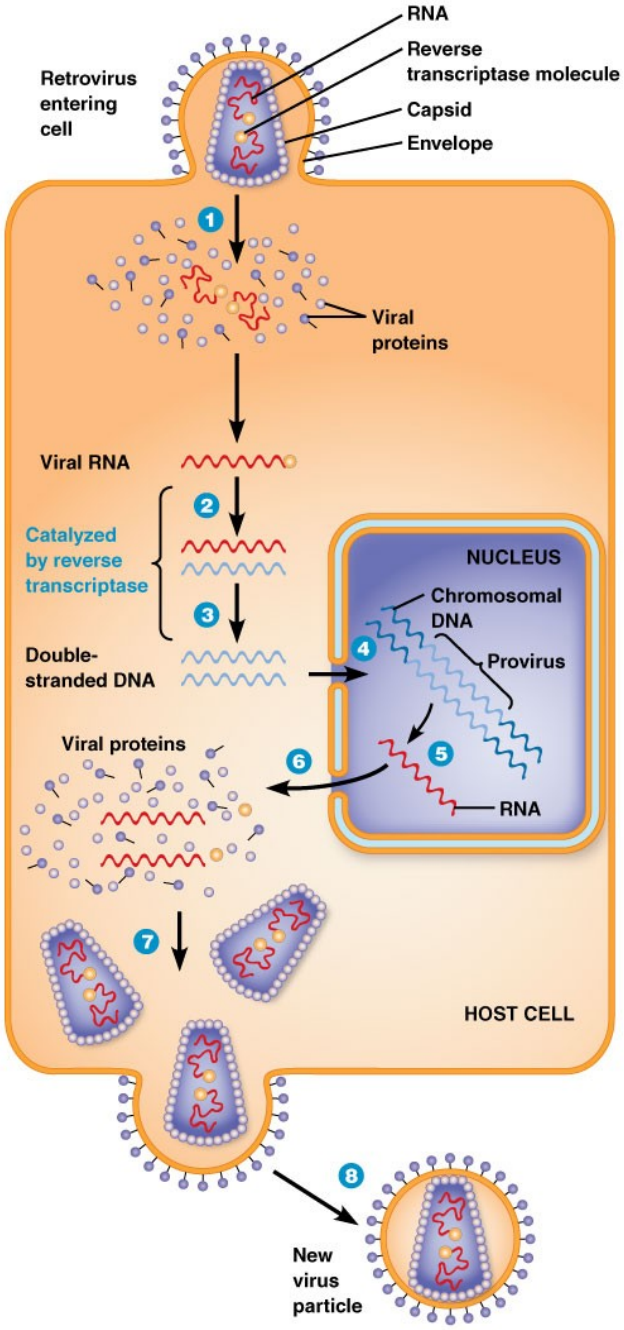
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Transcription of Retroviral DNA

- The reverse transcriptase then catalyzes the formation of a second DNA strand complementary to the first
- The resulting double-stranded DNA then enters the nucleus and integrates into the genome of the host
- The integrated viral genome is called a *provirus*
- It is replicated every time the host cell replicates its own DNA

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Transcription of Retroviral DNA

- Transcription of the proviral DNA produces RNA transcripts that function in two ways
- First, they serve as mRNA molecules that direct synthesis of viral proteins
- Second, some of these same transcripts (act as RNA genomes) are packaged with viral proteins into new virus particles
- The new viruses “bud” from the plasma membrane without necessarily killing the infected cell

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Recap

Genetic Code

Triplet, nonoverlapping, degenerate

64 codons → 20 amino acids

AUG = start; UAA, UAG, UGA = stop

Mutations

Point mutations: silent, missense, nonsense, nonstop

Indels → frameshift

Large mutations: duplication, inversion, translocation

Examples

Transcription

RNA pol reads template 3’→5’, synthesizes RNA 5’→3’

Template vs Coding strands

Steps: initiation → elongation → termination

Bacteria: coupled with translation

Eukaryotes: separated

Reverse Transcription

Viral RNA genome

Reverse transcriptase: RNA → DNA → dsDNA

Product integrates into host genome

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